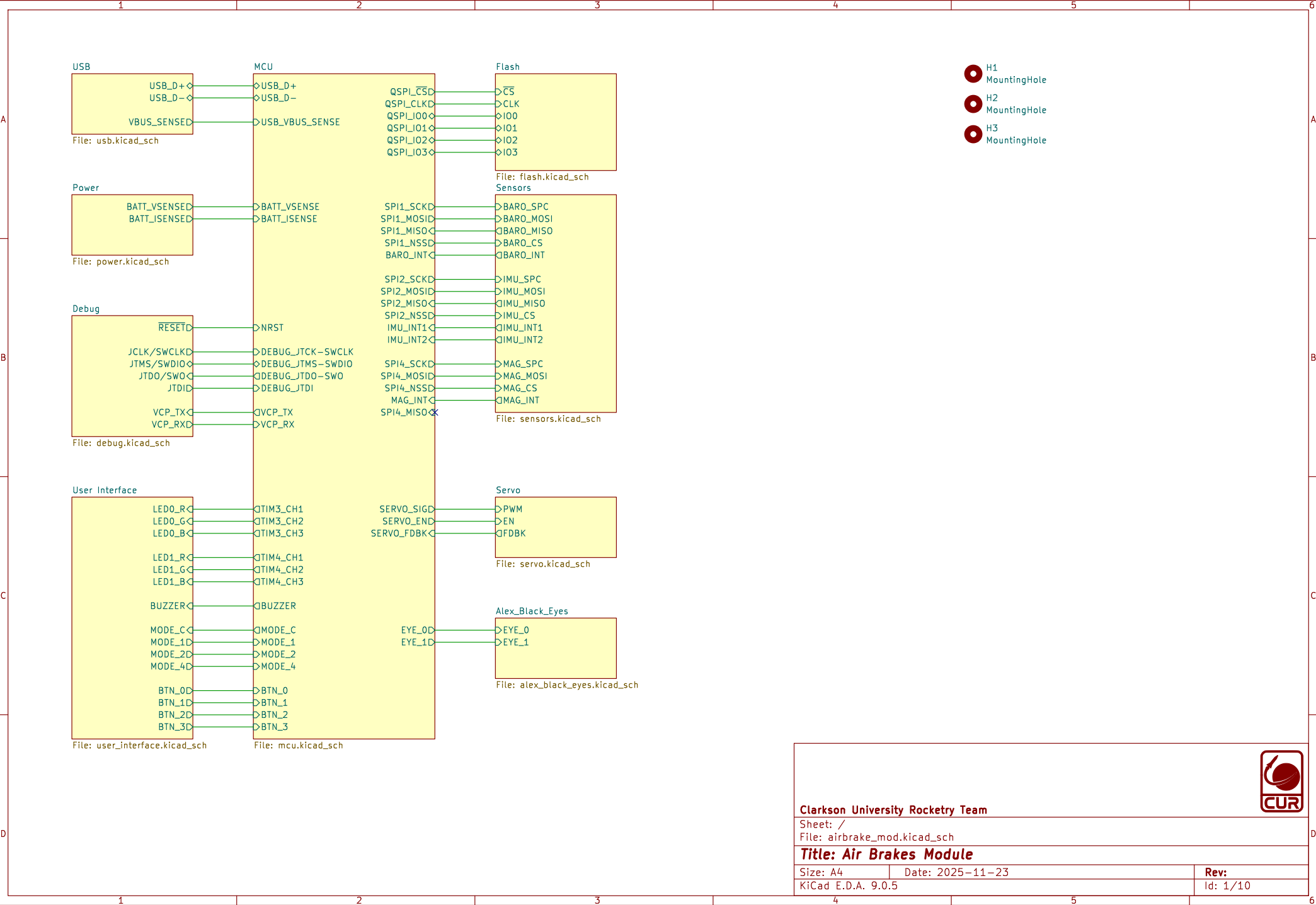
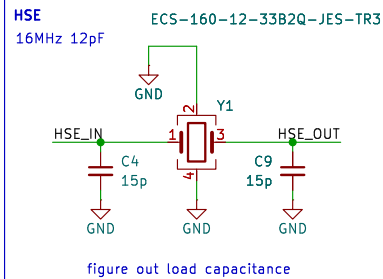
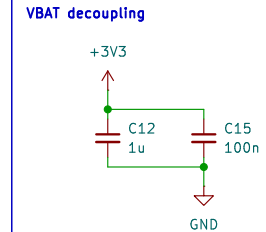
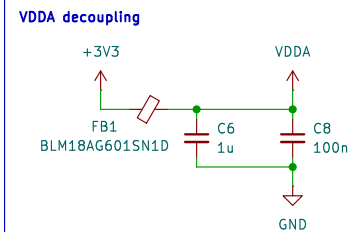
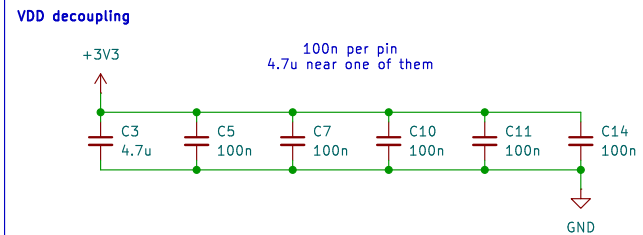
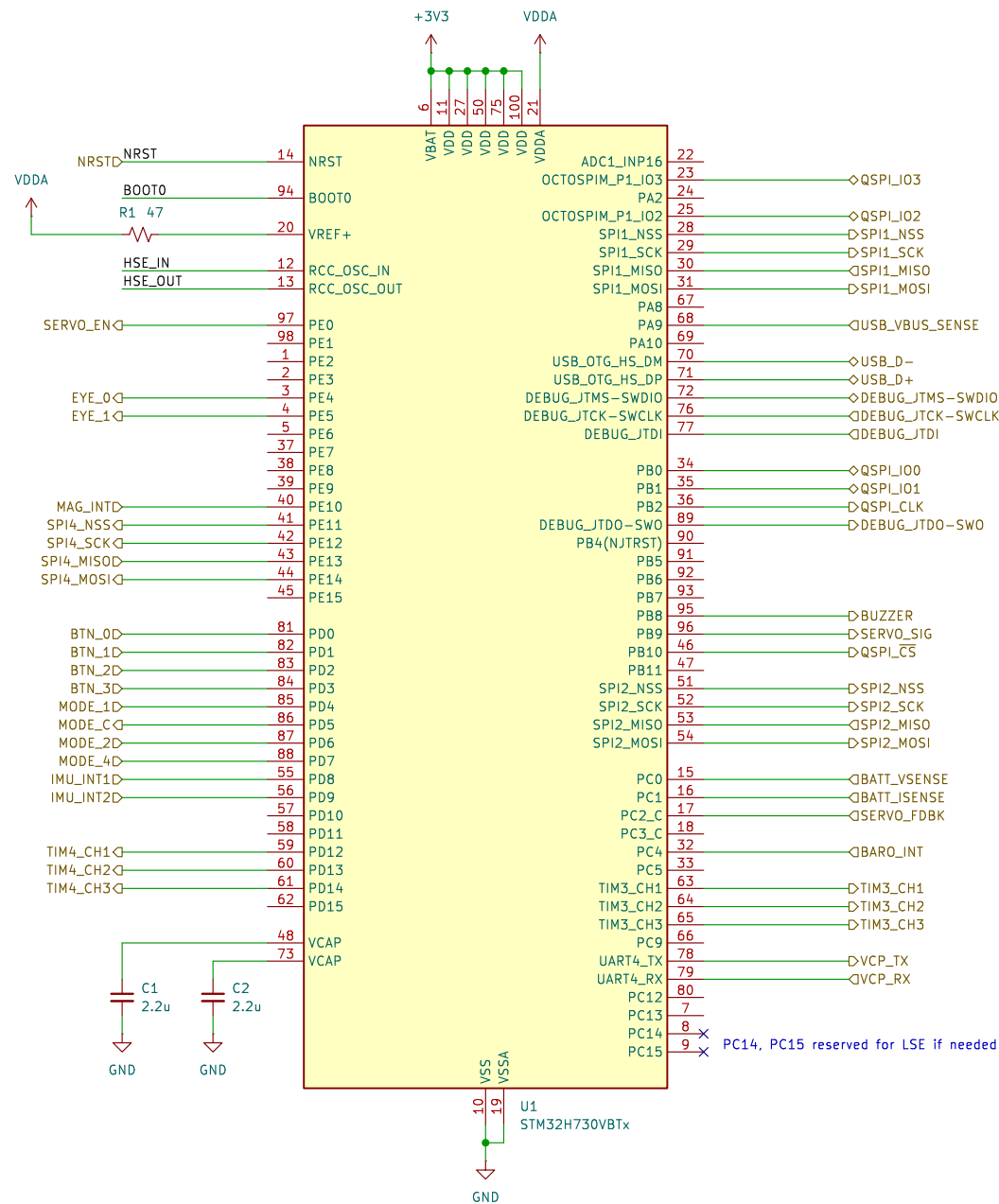
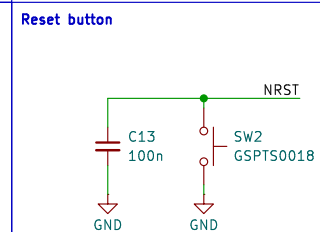
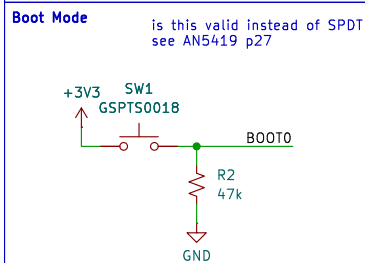






probably no LSE



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Sheet: /MCU/
File: mcu.kicad_sch

Title: Air Brakes Module

Size: A4	Date: 2025-11-23
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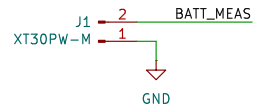
KiCad E.D.A. 9.0.5

Rev:

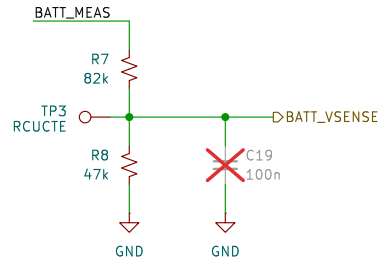
Id: 2/10

8	0.016	3.2	3971	0.128	greater than 1/8W
8.25	0.0165	3.3	4095	0.136125	max current sense
11	0.022	4	5460	0.242	max current less than 1/4W

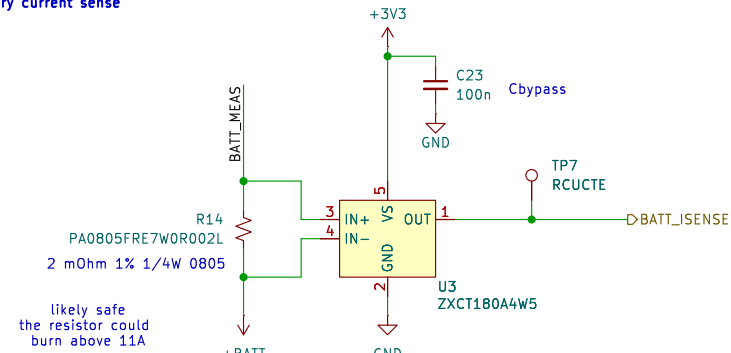
7.4V Lithium ion battery



Battery voltage measurement



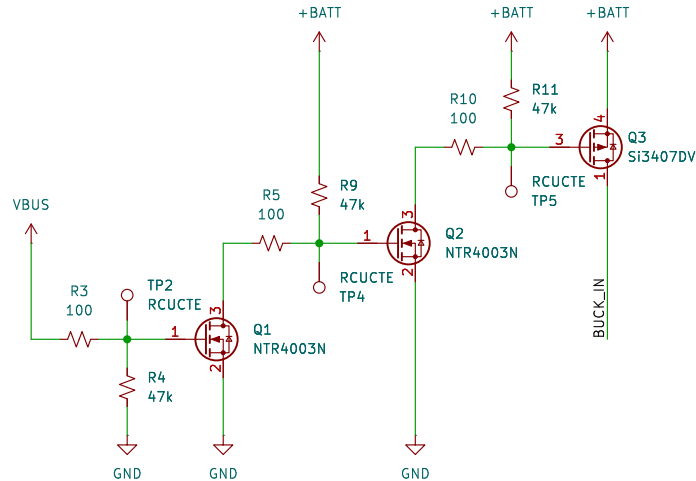
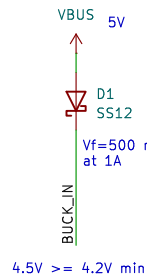
Battery current sense



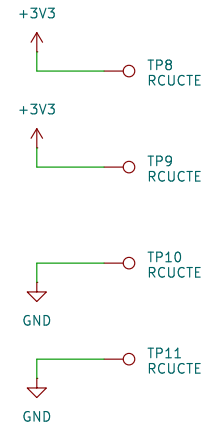
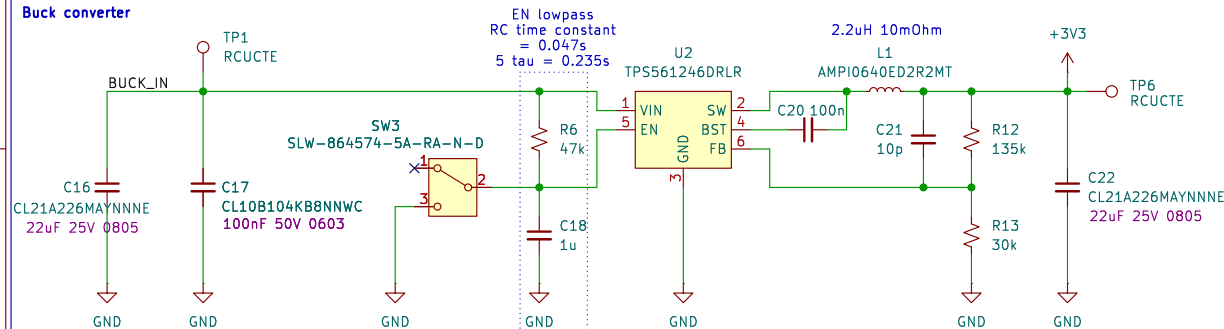
Battery power switch

PMOS ON when VBUS OFF

USB alt power in



Buck converter



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Sheet: /Power/

File: power.kicad_sch

Title: Air Brakes Module

Size: A4 Date: 2025-11-23

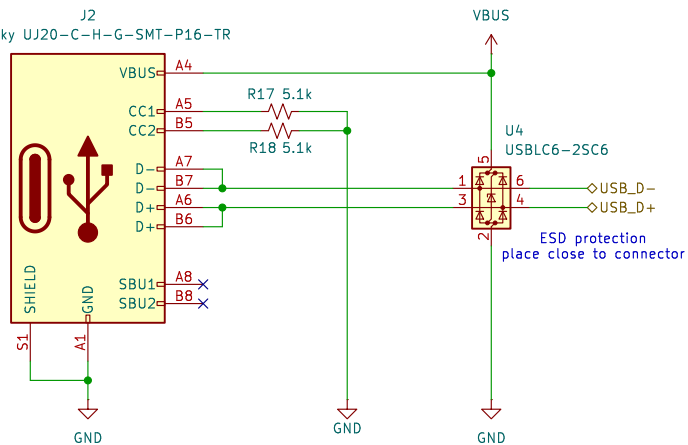
KiCad E.D.A. 9.0.5

Rev:

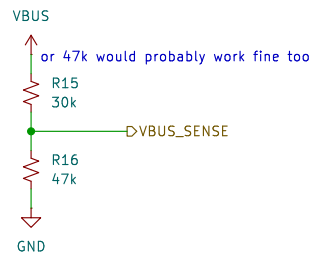
Id: 3/10

USB-C connector

J2
Same Sky UJ20-C-H-G-SMT-P16-TR

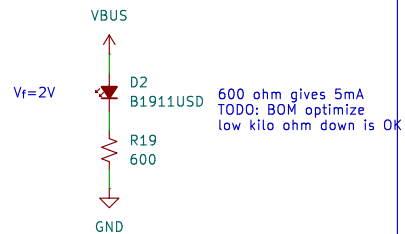


VBUS detect see AN4879 pg 13



VBUS_SENSE must be $\leq 4V$
to remain within absolute maximums

VBUS indicator



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Sheet: /USB/
File: usb.kicad_sch

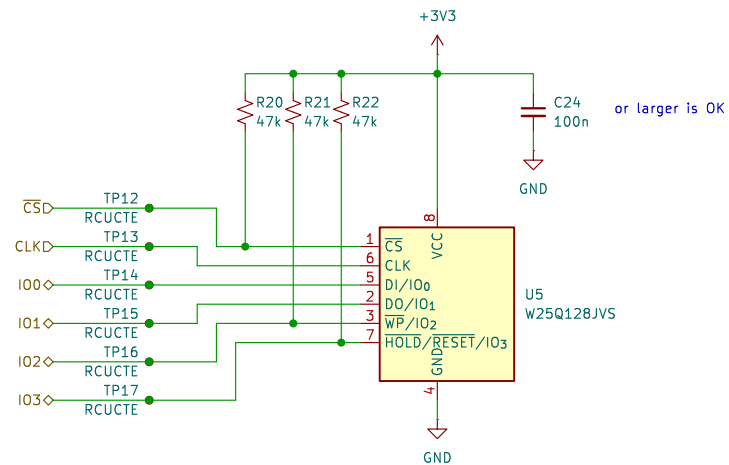
Title: Air Brakes Module

Size: A4 Date: 2025-11-23

KiCad E.D.A. 9.0.5

Rev:
Id: 4/10





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Sheet: /Flash/
File: flash.kicad_sch

Title: Air Brakes Module

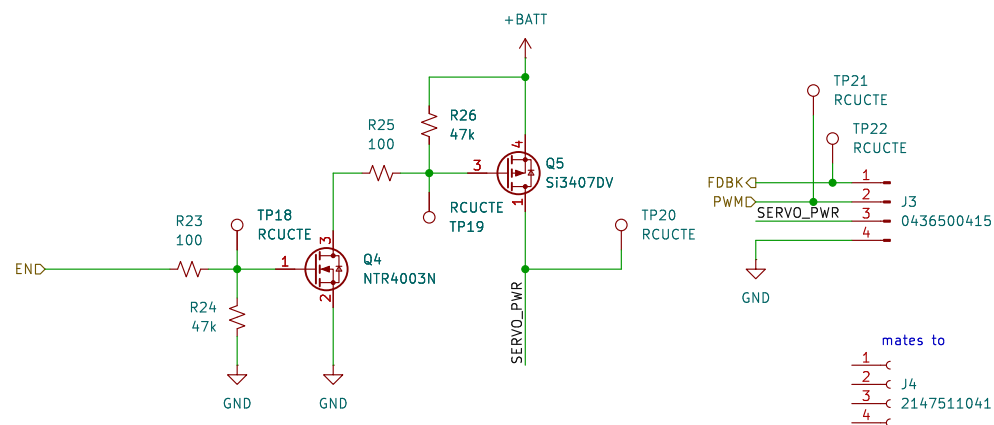
Size: A4

Date: 2025-11-23

Rev:

KiCad E.D.A. 9.0.5

Id: 5/10



Clarkson University Rocketry Team

Sheet: /Servo/
File: servo.kicad_sch

Title: Air Brakes Module

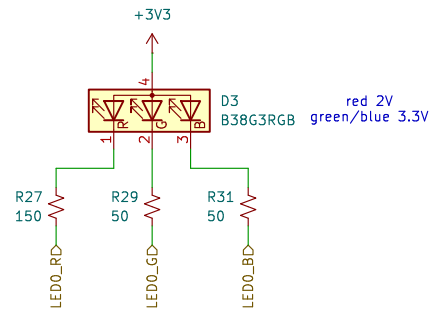
Size: A4
KiCad E.D.A. 9.0.5

Date: 2025-11-23

Rev:
Id: 6/10

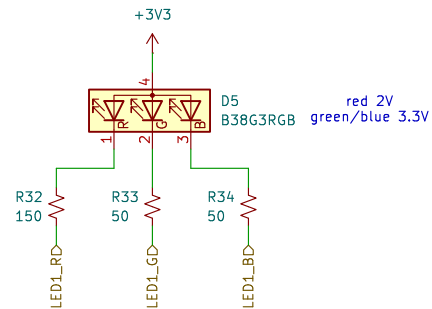
RGB LED Harvatek B38G3RGB

10 mA

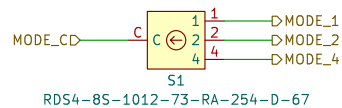


RGB LED Harvatek B38G3RGB

10 mA

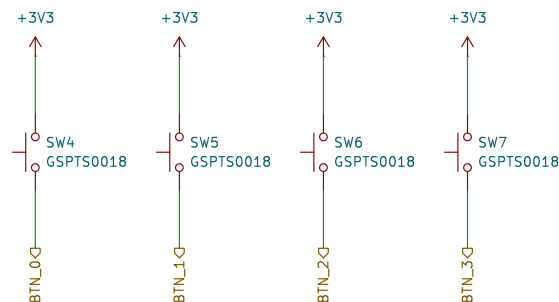


Mode switch

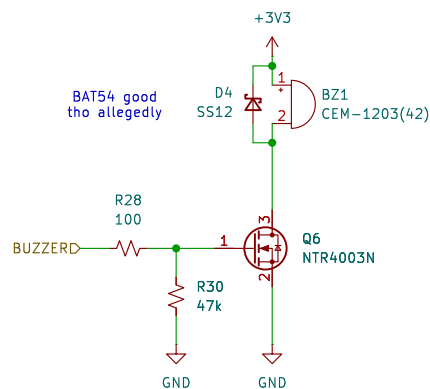


8 position BCD output
In this case, simply binary output

User buttons



Buzzer



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Sheet: /User Interface/
File: user_interface.kicad_sch

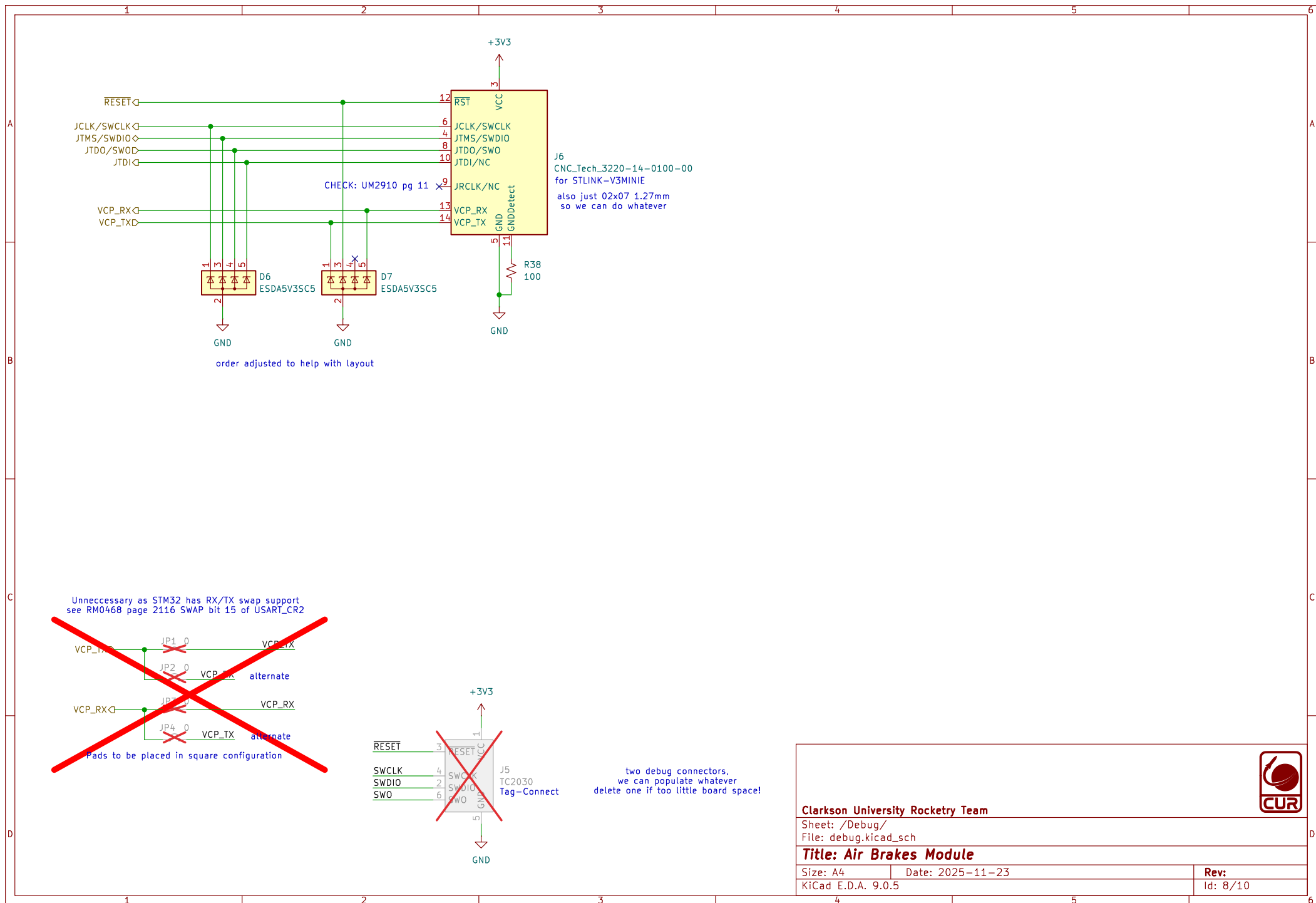
Title: Air Brakes Module

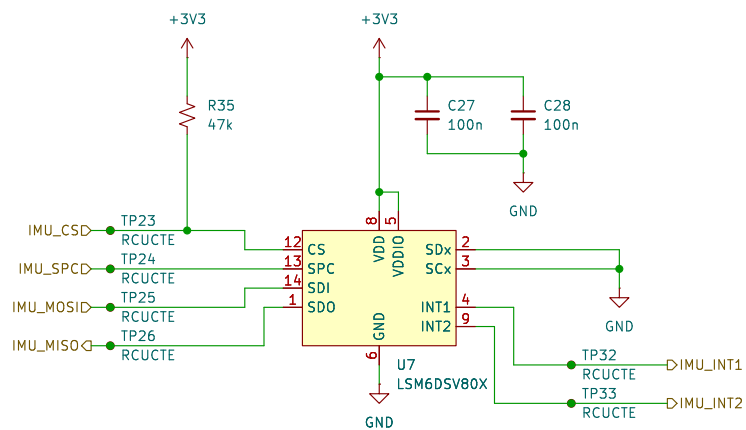
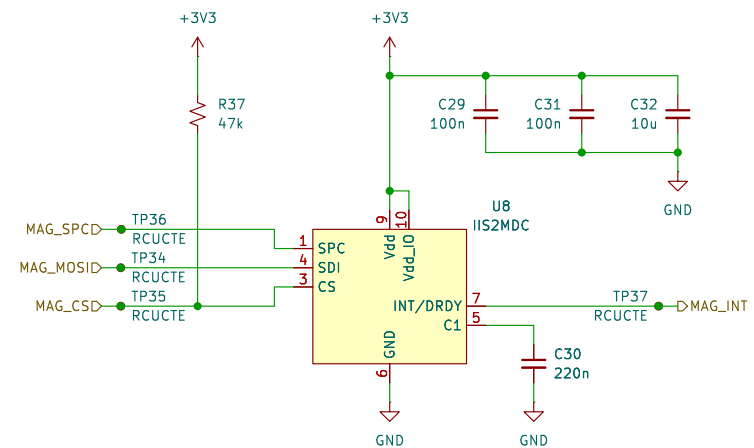
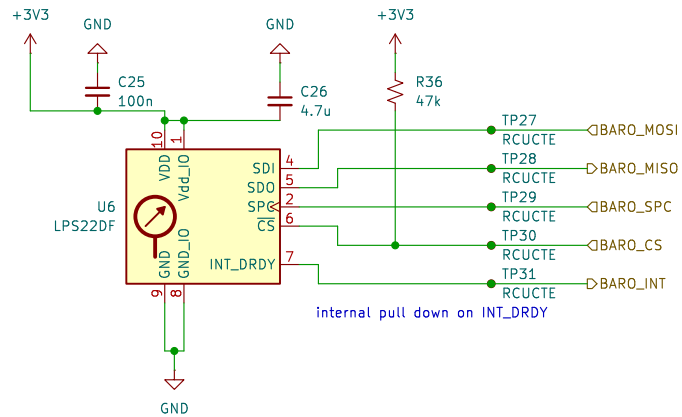
Size: A4 Date: 2025-11-23

KiCad E.D.A. 9.0.5

Rev:
Id: 7/10







MOSI/MISO swap can be achieved if needed
RM0468 page 2261 bit 15 I0SWP



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Sheet: /Sensors/
File: sensors.kicad_sch

Title: Air Brakes Module

Size: A4

Date: 2025-11-23

Rev:

KiCad E.D.A. 9.0.5

Id: 9/10

